

Document Control

- System Name: Application

Executive Summary

- This BRD outlines requirements for ingesting customer and order CSV data into Delta tables, cleansing data, computing total amounts, joining and applying SCD Type 2 for customer data, and aggregating daily customer spend.
- Implementation is specified using Delta Live Tables, leveraging defined names, schemas, and rules.

Business Objectives

- Load source customer and order CSVs into Delta tables.
- Cleanse source data by removing "Null" values and duplicates.
- Enrich order data with a computed TotalAmount column.
- Join customer and order data on CustId to create the ordersummary table.
- Apply SCD Type 2 change tracking for customer data.
- Create an aggregate table tracking customer spend by day.
- Implement the entire pipeline using Delta Live Tables.

In-Scope

- Source ingestion from defined paths into Delta tables.
- Creation and management of customer, order, ordersummary, and customeraggregatespend tables.
- Data cleansing actions (removal of nulls and duplicates).
- Computation of the TotalAmount column in order data.
- Join and SCD Type 2 handling for ordersummary.
- Aggregation of spend data into customeraggregatespend.
- Use of Delta Live Tables for pipeline execution.

Out-of-Scope

- Integration of data sources, systems, or fields not present in the requirements.
- Specification of security, access controls, SLAs, monitoring, and alerting.
- Performance tuning or scaling parameters.
- Scheduling, orchestration, or DevOps details outside of the pipeline steps.
- Data quality metrics beyond cleansing steps provided.
- User interfaces, reporting, or analytics layers.

Stakeholders

- Information not present in requirements input.

Business Requirements

BR-CORE-001 — Source Ingestion

- Description: Ingest customer and order CSV files from specified file system locations into respective Delta tables.
- Business Value: Provides a structured, persistent landing for raw business data.
- Priority: High

BR-CORE-002 — Data Schemas

- Description: Define Delta tables with specified schemas for customer (CustId, Name, EmailId, Region) and order (OrderId, ItemName, PricePerUnit, Qty, Date, CustId).
- Business Value: Ensures data structure consistency and mapping to business entities.
- Priority: High

BR-CORE-003 — Computed TotalAmount

- Description: Compute and add a TotalAmount column to order data ($\text{PricePerUnit} \times \text{Qty}$).
- Business Value: Enables direct financial analytics and aggregations.
- Priority: High

BR-CORE-004 — Data Cleansing

- Description: Remove "Null"/Null values and duplicate records from customer and order tables.
- Business Value: Improves data quality, ensuring trustworthy analytics and operations.
- Priority: High

BR-CORE-005 — Ordersummary Table Creation

- Description: Join cleansed customer and order data on CustId; create an ordersummary table applying SCD Type 2 for customer changes.
- Business Value: Provides an integrated, historical view of customer-order relationships.
- Priority: High

BR-CORE-006 — SCD Type 2 Change Handling

- Description: Implement SCD Type 2 logic for customer attributes in ordersummary, tracking full history via Active/Inactive flags and Start/End dates (fields referenced but not explicitly defined).
- Business Value: Enables accurate historical analysis and compliance.
- Priority: High

BR-CORE-007 — Aggregation Table Creation

- Description: Aggregate ordersummary data into customeraggregatespend, summing TotalAmount for each Name and Date.
- Business Value: Delivers actionable insights into customer spending patterns.
- Priority: High

BR-CORE-008 — Pipeline Implementation in Delta Live Tables

- Description: Orchestrate all data ingestion, transformation, SCD handling, and aggregation steps using Delta Live Tables.
- Business Value: Provides scalability, automation, and reliability for data processing workflows.
- Priority: High

Assumptions & Constraints

- Only explicitly defined content from requirements is assumed.
- Delta Live Tables is the mandated implementation technology.
- SCD Type 2 flag/date fields are referenced for process but not defined in schema.
- Any attributes not described in requirements are not considered.

Success Metrics

- All source and target tables created as specified, populated with expected data.
- Cleansing and computed logic executed with zero tolerance for data quality exceptions (as per rules specified).
- SCD Type 2 changes accurately reflected in ordersummary.
- Aggregated customer spend totals align with source data validation.
- End-to-end data pipeline successfully operationalized in Delta Live Tables.

Risks & Mitigations

- Ambiguity in SCD Type 2 tracking fields (StartDate, EndDate, Active): Clarify field requirements with stakeholders.
- Data quality not covered beyond null and duplicate removal: Plan for iterative enhancement as required.
- Lack of specified error handling, retries, and monitoring: Incorporate basic logging and alerting in the implementation phase.
- Platform/environment assumptions limited to use of Delta Live Tables: Engage technical architects for deployment requirements.

Approval

- Information not present in requirements input.

Appendix

Appendix A: Source Definitions

Source Name	Type	Location	Format	Notes
Customer source	CSV	/Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/customerdata	CSV	Loaded into Delta table customer
Order source	CSV	/Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/orderdata	CSV	Loaded into Delta table order

Appendix B: Target Definitions

	Layer	Table	Columns	Notes
	Raw	customer	CustId, Name, EmailId, Region	Target from
	Raw	order	OrderId, ItemName, PricePerUnit, Qty, Date, CustId	Target from
	Curated	ordersummary	CustId, Name, EmailId, Region, OrderId, ItemName, PricePerUnit, Qty, Date	SCD Type
pend	Aggregate	customeraggregatespend	Name, TotalAmount, Date	Aggregate

Appendix C: Transformations

Step	Transformation	Input	Output	Logic
1	Remove nulls	customer, order	cleansed tables	Delete records with Null/"Null"
2	Remove duplicates	customer, order	cleansed tables	Keep only distinct records
3	Compute TotalAmount	order	order	PricePerUnit × Qty
4	Join and SCD2	customer, order	ordersummary	Join on CustId; apply SCD Type 2
5	Aggregate	ordersummary	customeraggregatespend	sum(TotalAmount) by Name, Date

Appendix D: Business Rules

Rule ID	Area	Description	Source
BR-1	Computed Column	TotalAmount = PricePerUnit × Qty	Requirements
BR-2	Cleansing	Remove “Null” values	Requirements
BR-3	Deduplication	Remove duplicates	Requirements
BR-4	Join	Join customer to order on CustId	Requirements
BR-5	SCD Type 2	Inactivate/activate records, track StartDate/EndDate	Requirements
BR-6	Aggregation	Group by Name, Date; sum TotalAmount	Requirements

Appendix E: Process Flow

Step	Description	Dependency
1	Load customer and order CSVs	Source file system
2	Cleanse data (null/duplicate removal)	Step 1
3	Compute TotalAmount in order	Step 2

4	Join and apply SCD Type 2, build ordersummary	Step 3
5	Aggregate to customeraggregatespend	Step 4

Appendix F: Data Flow Diagram

```
```mermaid
```

```
flowchart TD
```

```
subgraph Ingestion
```

```
A["Customer source CSV"] --> B["Customer Delta Table"]
```

```
C["Order source CSV"] --> D["Order Delta Table"]
```

```
end
```

```
subgraph Cleansing
```

```
B --> B1["Customer Cleansed"]
```

```
D --> D1["Order Cleansed"]
```

```
end
```

```
subgraph Transformation
```

```
D1 --> D2["Add TotalAmount"]
```

```
end
```

```
subgraph Processing
```

```
B1 --> E["Ordersummary Join and SCD Type 2"]
```

```
D2 --> E
```

```
E --> F["Customer Aggregate Spend"]
```

```
end
```

```
```
```

[Mermaid render failed: Kroki render failed: 400]